

Honglin Cao

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Education

University of Waterloo, Bachelor of Computer Science

Sep 2020 – Aug 2025

- President's Scholarship of Distinction; Cumulative Average: **91%** (Math/CS Faculty: **93%**) / GPA: **4.0**

Experience

Engineer, Sys SW | NVIDIA Canada Dev. Inc. - Toronto, ON

Jun 2025 – Present

- Own the inference / serving platform powering production **LLM** deployments (**vLLM** / **SGLang**) end-to-end across **API**, catalog, and **Kubernetes** infrastructure; joined NVIDIA via its acquisition of **CentML**.
- Designed and shipped production inference features for **vLLM** / **SGLang**-backed deployments: **gRPC** inference with **Kubernetes-native** health checks, **HPA** autoscaling extended to **MIG** instances, **cluster GPU capacity** endpoints, a model-deployment recipe API integrated with **HuggingFace**, user-configurable **Prometheus** metrics endpoints, and per-deployment labels for **kube-state-metrics** cost attribution.
- Redesigned the **ArgoCD** / **Helm** parameter pipeline to render structured values directly, eliminating an entire class of string-templating bugs across every deployment chart.
- Shipped opt-in **session affinity** for multi-replica LLM serving; in production it raised a customer deployment's prefix / **KV-cache** hit rate from **55% to 96%**, cut mean **TTFT** by **74%** and **p99 latency** by **31%**, while *reducing* GPU utilization.
- Hardened the platform with **bearer-token** redaction, **owner-scoped** deployment mutations, **cross-org** endpoint-collision rejection, hardened user config-file mounts, organization-level **container registry** and **secrets vault** visibility, and remediation of critical **CVEs** across the API and image supply chain.
- Contributed upstream to **vLLM** (standard **Kubernetes-native gRPC** health checking) and to a **Rust** LLM inference gateway (health servicing and continuous **Watch** streaming).

Platform Software Engineer | CentML - Toronto, ON

Sep 2024 – Jun 2025

- Built the **Infrastructure-as-Code** local development environment in **Pulumi** / **Python**, mirroring production across **Kubernetes**, **Docker**, **Knative**, and **AWS** / **Google Cloud** with **LocalStack**, **Grafana**, **Prometheus**, **Istio**, and **Minikube** via **Helm** — closing billing, monitoring, deployment, and database-mocking gaps and **reducing integration-test execution time by over 90%**.
- Designed and implemented **APIs** for container deployment, billing, and user storage on **PostgreSQL**, with safe upgrade / downgrade paths via **SQLAlchemy** / **Alembic**; automated public **API-client** and non-root **container-image** releases via **GitHub Actions** with integration tests in **pytest**, **jwt**, and **WorkOS**.
- Contributed **3000+** lines to **open-source** projects (**llama-stack**, **huggingface_hub**, **litellm**, **huggingface.js**) in **Python** and **JavaScript**, alongside documentation providing alternative paths to interact with serverless endpoints.

Distributed Database Engineer (Co-op) | Huawei Canada - Markham, ON

2022 – 2023

- Designed an **RPC** sub-protocol over **TCP** / **RDMA** in **C** on multi-node **GaussDB** (removed packet-size limits, enabled **crash-recovery** messaging) and built its **unit-test** layer from scratch.
- Automated **TPC-C** / performance benchmarking (**perf**, **CPU flame graphs**, **jTPCC**) across **ARM** / **x86** via **GitLab CI** and **Jenkins**; standardized multi-node **GaussDB** benchmark templates.